

An Account of My Experiences With the Challenges Outside the Classroom: Mathematical Olympiad, Exhibition of Mathematical Instructional Aid, Pre-workshop Assignment, Poster contest

Ever since I started my career as a mathematics teacher educator, I was interested to know how the students react to mathematical challenges outside the classroom. I tried several experiments with different learning groups (see table 1). The challenge lies in making the target groups to think and develop mathematical reasoning beyond the classroom walls. I record my experiences here.

Experiment 1: Exhibition of Mathematical Instructional Aid

Preparation of instructional aids is important in a secondary teacher-training course in my ¹university. Weightage is given to innovative teaching aids while assessing the student teachers. I feel that putting mathematical ideas in projected and non-projected aids is a viable option to challenge the pre-service teachers, to make them materialise the mathematical ideas into teaching aids. This makes the pre-service teachers to perceive the mathematical concepts in different ways, which calls for creativity and divergent thinking.

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To give an idea of mathematical teaching aids to the prospective teachers, I have conducted one training cum exhibition program of mathematical instructional aids on 27.11.2000 and 28.11.2000 in ²Tuticorin, India. Twenty-five secondary mathematics student teachers participated in a week-long training session. Toward the final phase of the training program the teacher trainees prepared mathematical instructional aids and exhibited them. Seventy-five mathematics teachers and as many as seven hundred girl students were benefited from the exhibition.

Mathematics in nature, paper folding, tessellation, 3-D solid models like dodecahedron, Boolean operators, Pythagoras theorem, Angle of depression and elevation, similar triangles, mathematics and computer, digital adding machine, Queen of Mathematics, Mathematical tree, mathematical year 2000 are some of the ideas put forth by the student teachers in their instructional aids.

² a port town in Tamil Nadu, India - <http://www.mapsofindia.com/maps/tamilnadu/tamilnadu-district.htm>

Experiment 2: Model Mathematical Olympiad

I carried out an innovative experiment called *Model Mathematical Olympiad* (MMO) designed to the goal of strengthening the mathematical capabilities of exceptionally talented Year 9 students from Tuticorin conducted in the year 2002. This experiment addresses the issues involved in promoting young mathematicians in Tuticorin who have otherwise no access to such mathematical contests. The issues have been identified in terms of mathematics curricula followed in different schools, location of the schools, performance by sex, type of schools, and medium of instruction on a pilot basis.

Seventy-five students (Year 9) and forty-five teachers from forty-six schools participated in the MMO. Students were directed to solve fifteen mathematical problems. Students' performance was not affected by sex factor but medium of language had an effect on students' scores.

The need to educate the teachers about Olympiads and mobilising funds to conduct such programs on yearly basis especially in the rural areas is realised through this experiment.

Experiment 3: Pre-workshop Assignment

A workshop on developing problem solving competency was planned for the secondary student teachers on 9.11.2002 and 10.11.2002 in Tuticorin, India. Before the workshop, each teacher trainee was asked to solve five sets of problems (eleven problems) to be discussed during the workshop. Twenty secondary mathematics teachers (F =12 and M=8) participated in the workshop. The problems were sent to the student teachers by post beforehand and the participants were requested to come with the solution procedures. It was expected that these solution processes would form a basis for meaningful participation during the course of workshop. It was found that almost all the participants turned up with different solution procedures. This was helpful in identifying the problem solving strategies (such as Clarifying, Visualizing, Analogy, Sub goal, Algebraic, Brain-Storming, Questioning, Unit Analysis, Identifying, Team, Trial-Error, Front End Entry Points, Rival Hypotheses, Induction- Find a Pattern, Heuristics, Algorithms, Work Backwards, Look Back, Incubate, and Go for Help) preferred by pre-service teachers by way of analyzing their trait thinking behaviour.

Experiment 4: Poster Contest

A poster contest 'Mathematics-Language of the Universe' was conducted for student teachers on 10.11.2002 in Tuticorin, India. Twenty student teachers (F =12 and M=8) from seven colleges of education in India participated in the poster contest. I designed the poster contest to meet the objective of out bringing the concept orientation of student teachers towards mathematics and to provide an opportunity to organise their thinking about mathematics in a creative way. The contest delineates the specific objectives of exposing the significance of mathematics, utility value of mathematics in daily life and social values of mathematics by way of eliciting the prospective math teacher's ideas.

Philosophy in Mathematics, aesthetic beauty of Mathematics, Mathematics and art of living, Mathematics and religious tolerance are some of the themes selected by the student teachers. Role of Mathematics in the development of human science, Mathematical symbols and shapes used in daily life, Mathematics and aeronautics are some other concepts used by the contestants. Four student teachers depicted how Mathematics promotes other subjects by choosing different personification of Mathematics such as the rising sun, tree root, the Sun with planets and peacock with feathers. One contestant tried linking mathematics with the structure of chemical molecules. This contest served as a window to analyse the beliefs of future teachers about Mathematics as a subject.

Table 1: Experiments

Experiments	Place and Date	Target Group	Driving Questions	Aim	Goals Attained	Further Development
Mathematical Instructional Aid	Tuticorin, India 27.11.2000	Pre-service secondary mathematics teachers (N=25)	How pre-service teachers conceptualise mathematical ideas in the form of instructional aids?	To provide training and to provide an opportunity to bring out the creative thinking of pre- service teachers	Pre-service teachers worked individually on different mathematical ideas and displayed their products. As many as seventy-five mathematics teachers and seven hundred girl students viewed the exhibited materials.	Modeling mathematical ideas to promote divergent thinking. Same mathematical concept is conceptualized in different themes.

Experiments	Place and Date	Target Group	Driving Questions	Aim	Goals Attained	Further Development
Model Mathematical Olympiad	Tuticorin, India 19.1.2002	Exceptionally talented Year 9 students (N=75; G=32 and B=43)	What models exist for delivering mathematical challenges, especially to students in rural or isolated environments? Have these approaches been successful? What challenges exist?	To provide an opportunity for the students to exhibit their mathematical excellence	Created awareness, among rural students and teachers about mathematical Olympiad	Follow-up and sustained efforts to conduct such programs to benefit rural students

Experiments	Place and Date	Target Group	Driving Questions	Aim	Goals Attained	Further Development
Pre-workshop Assignment	Tuticorin, India 9.11.2002	Pre-service secondary mathematics teachers (N=20; F =12 and M=8)	In what way we can orient the pre-service secondary teachers towards the better participation in a problem-solving workshop?	To provide a platform for discussion on problem-solving techniques	Pre-assignment problems helped the pre-service teachers to think about various problem solving techniques beforehand	Quality of mathematical reasoning to a given set of problems could be focused

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Poster contest	Tuticorin, India 10.11.2002	Pre-service secondary mathematics teachers (N=20; F =12 and M=8)	What comes to the mind of pre-service secondary mathematics teachers' when they think of mathematics as the language of universe?	To analyse pre- service secondary mathematics teachers' beliefs about mathematics as the language of universe	Significance of mathematics, utility value of mathematics in daily life and social values of mathematics are brought out by eliciting prospective math teacher's ideas.	Student teachers' beliefs about mathematics can be explored in a similar fashion